13) $u = \dfrac{12i + 5j}{\sqrt{12^2+5^2}} = \dfrac{12i}{13} + \dfrac{5}{13}j$

$$\nabla g = \left(\frac{y^2+2}{(xy+2)^2}\right)i + \left(\frac{x^2+2}{(xy+2)^2}\right)j = 3i - 3j$$

$$(D_u g) = 3 \cdot \frac{12i}{13} - 3 \cdot \frac{5}{13}j = \frac{36}{13} - \frac{15}{13} = \frac{21}{13}$$